

Organic Chemistry

7



Reference: Coursebook - IIT Foundation Chemistry Class 9; Chapter - Organic Chemistry; Page number - 7.1–7.26

Assessment Test I

Time: 30 min.

Directions for questions from 1 to 15: Select the correct answer from the given options.

Space for rough work

1. The number of sigma and pi bonds in 4-methyl 2-hexyne are _____ and _____, respectively.

(a) 18 and 2 (b) 17 and 1
(c) 17 and 2 (d) 18 and 1

2. Among the following, identify the formula of a compound which can exhibit both chain and position isomerism.

(a) C_5H_{12} (b) C_4H_{10} (c) C_4H_6 (d) C_6H_{14}

3. Among the following compounds, identify the compound which possesses similar root word and position of substituent as 3-methyl 1-pentene.

(a) $C-C-C-C-C-C$ (b) $C-C=C-C-C$
 $\quad \quad \quad |$ $\quad \quad \quad |$
 $\quad \quad \quad C-C$ $\quad \quad \quad C-C$
 (c) $C-C-C-C-C$ (d) $C-C$
 $\quad \quad \quad |$ $\quad \quad \quad |$
 $\quad \quad \quad C$ $\quad \quad \quad C-C-C-C$

4. Arrange the following compounds in the increasing order of root word.

(A) $C-C-C-C-C=C$
 $\quad \quad \quad \quad \quad |$
 $\quad \quad \quad \quad \quad C-C-C$
 (B) $C-C-C-C-C$
 $\quad \quad \quad |$
 $\quad \quad \quad C=C$
 $\quad \quad \quad |$
 $\quad \quad \quad C$
 (C) $C-C-C-C$
 $\quad \quad \quad |$
 $\quad \quad \quad C-C$
 (D) $C-C-C=C$
 $\quad \quad \quad |$
 $\quad \quad \quad C=C$
 (E) $C-C-C-C-C-C$
 $\quad \quad \quad |$
 $\quad \quad \quad C-C-C-C-C-C-C$

- (a) BCDAE (b) DEBAC
(c) DCBAE (d) CBEAD

5. Arrange the following main fractions of refining of petroleum in the correct order of temperature range in which they are obtained.

- (A) Lubricating oil (B) Kerosene oil
(C) Petroleum gas (D) Paraffin wax
(E) Fuel oil

- (a) CBDAE (b) DEBAC
(c) DCBAE (d) CBEAD

6. **Assertion (A):** Bromine water is used to distinguish between ethane and ethene.

Reason (R): Ethene undergoes addition reaction with bromine water.

- (a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

7. **Assertion (A):** Ethyne decolourises pink colour of alkaline KMnO_4 solution.

Reason (R): Ethyne gets oxidized to glyoxal.

- (a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

8. Match the entries of Column A with those of Column B.

Column A (Main Fraction of Distribution of Coal Tar)	Column B (Composition)
(i) Green oil	(A) Pitch
(ii) Heavy oil	(B) Benzene
(iii) Residue	(C) Anthracene
(iv) Light oil	(D) Phenol

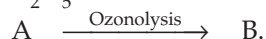
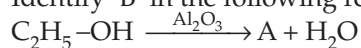
- (a) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (B)
(b) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (B); (iv) $\rightarrow$ (A)
(c) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (C)
(d) (i) $\rightarrow$ (D); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (B); (iv) $\rightarrow$ (C)

9. The molecular mass of a hydrocarbon 'X' is 70. What would be the molecular formulae of its nearest neighbouring homologues?

- (a) C_4H_8 and C_6H_{12} (b) C_4H_{10} and C_6H_{14}
(c) C_4H_6 and C_6H_{10} (d) C_5H_{12} and C_7H_{14}

Space for rough work

10. Identify 'B' in the following reaction



- (a) Glyoxal (b) Formaldehyde
(c) Oxalic acid (d) Acetaldehyde
11. 1 mole of hydrocarbon 'X' on complete combustion with 3 moles of O_2 gives 2 moles of CO_2 and 2 moles of water. Then, X can be prepared by
- (a) decarboxylation of sodium acetate.
(b) dehydration of ethyl alcohol in the presence of conc. H_2SO_4 at 180°C .
(c) hydrolysis of calcium carbide.
(d) heating iodoform with silver powder.
12. What would be the pair of products formed by treating alkyne with the least number of carbon atoms with bromine water?
- (a) Dibromoethane, tetrabromoethane
(b) Dibromoethene, tetrabromoethene
(c) Dibromoethane, tetrabromoethene
(d) Dibromoethene, tetrabromoethane.
13. In which among the following reactions does the total number of bonds remains the same?
- (a) Chlorination of methane (b) Hydrogenation of ethene
(c) Dehydration of ethyl alcohol (d) Hydrolysis of ethyne
14. Saturated hydrocarbon with vapour density 8 can be prepared by _____.
- (a) dehydration of alcohol
(b) decarboxylation of sodium acetate
(c) hydrolysis of calcium carbide
(d) hydrogenation of ethene
15. Identify the functional isomers among the following.
- (a) Pentanol and dimethyl ether (b) Propanol and dimethyl ether
(c) Butanol and diethyl ether (d) Ethanol and diethyl ether

Assessment Test II

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

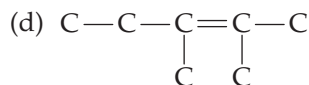
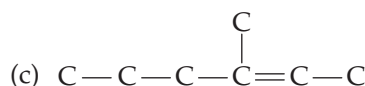
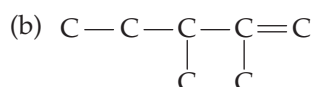
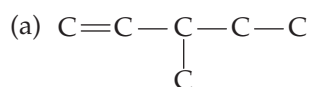
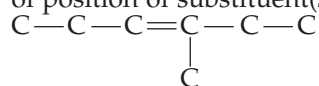
1. Identify the pair of compounds which possesses the same total number of carbon-carbon bonds.

(a) Benzene, toluene (b) Toluene, aniline
(c) Benzene, benzoic acid (d) Toluene, benzoic acid

2. 3-methyl 1-butene is one of the position isomers of compound 'X'. Then, identify the chain isomer.

(a) 2-methyl 2-butene (b) 2-methyl 1-butene
(c) 3-methyl 1-butene (d) 1-pentene

3. Among the following compounds, identify the compound in which the sum of position of substituent(s) and position of unsaturation is equal to that in



4. Arrange the following compounds in the increasing order of total number of carbon atoms present in the compound.

(A) 3, 4, 5-trimethyl heptane (B) 3, 3-diethyl pentane
(C) 5-ethyl nonene (D) 2-ethyl pentane
(E) 4-methyl 1-heptene

(a) BCDAE (b) DEBAC
(c) DCBAE (d) CBEAD

5. Arrange the compositions given below in the order of fuel oil, paraffin wax, crude naphtha, kerosene oil and lubricating oil.

(A) $\text{C}_5 - \text{C}_{10}$ (B) $\text{C}_{10} - \text{C}_{16}$
(C) $\text{C}_{16} - \text{C}_{20}$ (D) $\text{C}_{20} - \text{C}_{24}$
(E) Above C_{24}

(a) CBEAD (b) CEABD
(c) CEBDA (d) CEADB

6. **Assertion (A):** Ammoniacal silver nitrate solution is used to distinguish ethene from ethyne.

Reason (R): Ethyne undergoes addition reaction with ammoniacal silver nitrate solution.

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true and R is false.
- (d) A is false and R is true.

7. **Assertion (A):** Ethene and ethyne can be distinguished by alkaline KMnO_4 solution.

Reason (R): Alkaline KMnO_4 solution is a good oxidizing agent.

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true and R is false.
- (d) A is false and R is true.

8. Match the entries of Column A with those of Column B.

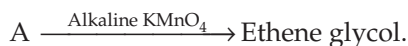
Column A	Column B
(i) Middle oil	(A) $> 400^\circ\text{C}$
(ii) Heavy oil	(B) $170^\circ\text{C} - 230^\circ\text{C}$
(iii) Green oil	(C) $230^\circ\text{C} - 270^\circ\text{C}$
(iv) Residue	(D) $270^\circ\text{C} - 400^\circ\text{C}$

- (a) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (C)
- (b) (i) $\rightarrow$ (D); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (B); (iv) $\rightarrow$ (A)
- (c) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (B)
- (d) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (A)

9. Two homologues of a series differ by 56 in their molecular mass. What could be their molecular formulae?

- (a) C_6H_{14} and C_9H_{20}
- (b) C_2H_6 and C_6H_{12}
- (c) C_4H_8 and C_8H_{14}
- (d) C_4H_{10} and C_8H_{18}

10. In the following reaction, compound A, on treatment with water, gives B. What could be A and B?



- (a) A $\rightarrow$ Ethene, B $\rightarrow$ Acetaldehyde
- (b) A $\rightarrow$ Ethyne, B $\rightarrow$ Ethyl alcohol
- (c) A $\rightarrow$ Ethene, B $\rightarrow$ Ethyl alcohol
- (d) A $\rightarrow$ Ethyne, B $\rightarrow$ Acetaldehyde

11. Alkane 'X' with six carbon atoms on cracking gives alkane 'Y' and alkene 'Z'. Identify the correct statement regarding Z.

- (a) Z on polymerisation gives benzene.
- (b) Z gives red precipitate with ammoniacal cuprous chloride.
- (c) Z can be prepared by dehydration of ethyl alcohol in the presence of Al_2O_3 at 170°C .
- (d) Z is used in oxyacetylene flame.

Space for rough work

12. The ratio of the number of carbon atoms to hydrogen atoms is 1:1 in an alkyne 'X'. X on ozonolysis gives _____.
(a) oxalic acid (b) glyoxal
(c) formaldehyde (d) acetaldehyde
13. In which among the following reactions does the total number of bonds increases by one?
(a) Complete hydrogenation of ethyne
(b) Hydrogenation of ethene
(c) Chlorination of methyl chloride
(d) Oxidation of ethene
14. Hydrocarbon 'X' requires 4 moles of chlorine gas for complete chlorination. Identify the true statement regarding X.
(a) On pyrolysis, it gives C and H₂.
(b) On reaction with steam, it gives CO₂ and H₂.
(c) It is obtained by hydrogenation of ethyne.
(d) Its vapour density is 16.
15. Alcohol 'X' and diethyl ether are functional isomers. Identify true statement regarding X.
(a) It has three chain isomers.
(b) It has two position isomers.
(c) It is the third member of the homologous series.
(d) Its root word is pent.

Assessment Test III

Time: 30 min.

Space for rough work

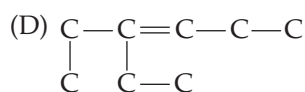
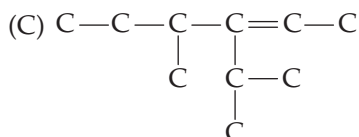
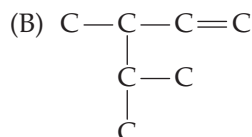
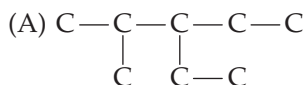
Directions for questions from 1 to 15: Select the correct answer from the given options.

1. **Assertion (A):** Compound with molecular formula C_4H_8 decolorises bromine water.

Reason (R): C_4H_8 is an unsaturated compound.

- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true and R is false.
 (d) A is false and R is true.
2. **Assertion (A):** Natural gas is used in the manufacture of fertilizers.
Reason (R): The major component of natural gas gives hydrogen gas on pyrolysis.
- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true and R is false.
 (d) A is false and R is true.

3. Arrange the following organic compounds in the increasing order of the root word.



- (a) BDAC (b) BDCA (c) DBCA (d) DBAC
4. The products formed by ethylene in different reactions are given below.
 Ethyl alcohol, formaldehyde, ethylene glycol and carbon dioxide
 Arrange the following reactions/reagents in the order of the above products.
 (A) Combustion (B) Addition of water
 (C) Oxidation with alkaline $KMnO_4$ (D) Ozonolysis
 (a) DBAC (b) BDAC (c) BDCA (d) DBCA
5. Match the entries of Column A with those of Column B.

Column A (Main Fraction)	Column B (Temperature Range)
(i) Crude naphtha	(A) $350^\circ\text{C} - 400^\circ\text{C}$
(ii) Kerosene oil	(B) $30^\circ\text{C} - 150^\circ\text{C}$
(iii) Fuel oil	(C) $150^\circ\text{C} - 300^\circ\text{C}$
(iv) Lubricating oil	(D) $300^\circ\text{C} - 350^\circ\text{C}$

- (a) (i) $\rightarrow$ (D); (ii) $\rightarrow$ (B); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (C)
(b) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (C)
(c) (i) $\rightarrow$ (D); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (B); (iv) $\rightarrow$ (A)
(d) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (A)
6. The vapour density of a saturated aliphatic hydrocarbon 'X' is 36. Determine the number of chain isomers of X.
(a) 5 (b) 4 (c) 2 (d) 3
7. The position isomer of 2-methyl pentane is _____.
(a) 3-methyl pentane (b) 2, 3-dimethyl butane
(c) 2, 2-dimethyl butane (d) *n*-pentane
8. Which among the following formulae represents saturated hydrocarbon as well as unsaturated hydrocarbon?
(a) C_4H_8 (b) C_5H_8 (c) C_6H_{14} (d) C_7H_{16}
9. The carbide of a metal on hydrolysis gives hydrocarbon 'X' with a characteristic garlic odour. Determine the number of moles of oxygen required for complete combustion of one mole of X.
(a) 2.5 (b) 2 (c) 3.5 (d) 3
10. Which among the following pairs does not belong to the same homologous series?
(a) C_2H_6 , C_4H_{10} (b) C_4H_8 , C_5H_{10}
(c) C_5H_8 , C_7H_{12} (d) C_3H_8 , C_4H_6
11. The role of sunlight in the chlorination of methane is _____.
(a) to break chlorine molecules
(b) to break the bond between carbon and hydrogen in CH_4 molecule
(c) to form a bond between carbon and chlorine
(d) to form a bond between hydrogen and chlorine
12. Methyl iodide on reduction with zinc-copper couple, in the presence of ethyl alcohol, gives 'X', which on treatment with steam in the presence of Al_2O_3 and temperature at $750^\circ C$ gives 'Y' and 'Z'. Identify X, Y and Z.
(a) $X \rightarrow CH_4$, $Y \rightarrow CO$, $Z \rightarrow H_2$
(b) $X \rightarrow C_2H_4$, $Y \rightarrow CO_2$, $Z \rightarrow H_2$
(c) $X \rightarrow CH_4$, $Y \rightarrow CO_2$, $Z \rightarrow H_2$
(d) $X \rightarrow C_2H_4$, $Y \rightarrow CO$, $Z \rightarrow H_2$
13. Hydrocarbon 'A' can be prepared by heating iodoform with silver powder. Identify the wrong statement regarding A.
(a) A is an unsaturated hydrocarbon with 2 carbon atoms.
(b) A forms silver acetylide with ammoniacal silver nitrate.
(c) A decolourises bromine water.
(d) A, on polymerization, gives polythene.

14. The major component of light oil is _____.
(a) phenol (b) pitch
(c) benzene (d) naphthalene
15. The molar ratio of CO_2 and H_2O produced by the complete combustion of one mole of hydrocarbon 'X' is 4:3. Determine the number of possible position isomers of X.
(a) 4 (b) 3 (c) 2 (d) 1

Space for rough work

Assessment Test IV

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. **Assertion (A):** The tendency of vanaspati to get rancid is less than corresponding vegetable oil.

Reason (R): Unsaturation is more in vanaspati.

- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true and R is false.
 (d) A is false and R is true.

2. **Assertion (A):** CNG is a better fuel than LPG.

Reason (R): The major component of CNG is methane and that of LPG are propane and butane.

- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true and R is false.
 (d) A is false and R is true.

3. Arrange the following compounds in the decreasing order of the total number of hydrogen atoms present in them.

- (A) 4-methyl 2-pentyne (B) 4, 4-dimethyl 2-hexene
 (C) 3, 4, 5-trimethyl 2-hexene (D) 2, 3-dimethyl 2-pentene
 (a) CBDA (b) BDAC (c) CBAD (d) BCDA

4. The products formed by ethyne in different reactions are given below.

Glyoxal, oxalic acid, benzene and acetaldehyde

Arrange the following reactions in the order of the above products.

- (A) Polymerization
 (B) Oxidation with alkaline KMnO_4
 (C) Ozonolysis
 (D) Addition of water
 (a) DCBA (b) CBAD (c) CDBA (d) DCAB

5. Match the entries of Column A with those of Column B.

Column A (Main Fraction)	Column B (Composition)
(i) Fuel oil	(A) $\text{C}_{20} - \text{C}_{24}$
(ii) Crude naphtha	(B) $\text{C}_{16} - \text{C}_{20}$
(iii) Lubricating oil	(C) $\text{C}_{10} - \text{C}_{16}$
(iv) Kerosene	(D) $\text{C}_5 - \text{C}_{10}$